DSP · L10

IIR design: bilinear transform and Butterworth

Practice questions · Day 19 · Wed 11 Nov · pack D11

5. Practice

From your course files

T9 Q4 (tutorial). Given the following digital filter specifications, obtain the corresponding analogue filter specifications for bilinear transformation:

$$0.8 \leq |H(\omega)| \leq 1, \quad 0 \leq \omega \leq 0.3\pi$$

$$|H(\omega)| \leq 0.1, \quad 0.6\pi \leq \omega \leq \pi.$$

T9 Q2 (tutorial). Convert the analogue filter

$$H(s) = \frac{s + 0.1}{(s + 0.1)^2 + 9}$$

into a digital IIR filter using the bilinear transformation.

T9 Q3 (tutorial). An ideal analogue integrator is described by $H_a(s) = 1/s$.

(a) Convert the analogue integrator into a digital integrator using the bilinear transformation. Write the difference equation for the digital integrator.
(b) Roughly sketch the magnitude and phase response of the analogue integrator.
(c) Obtain the frequency response of the digital integrator $H(\omega)$, and roughly sketch the magnitude and phase responses.
(d) Compare the magnitude and phase responses in parts (b) and (c), and comment on how well they match.
(e) Considering that the digital integrator has a pole at $z = 1$, what restrictions may need to be placed on the input signal?

T9 Q6 (tutorial). The poles of an analogue Butterworth filter are located on a circle of radius 50 in the s-plane. Find the filter order N if the minimum attenuation of the filter is 80 dB at 100 Hz.

T9 Q5 (tutorial). A digital lowpass filter is required to have the following specifications:

Passband ripple: ≤ 1 dB
Stopband attenuation: ≥ 20 dB
Passband edge frequency: 3 kHz
Stopband edge frequency: 9 kHz
Sampling frequency: 30 kHz

Obtain the transfer function and difference equation for the digital filter by applying the bilinear transformation to an analogue Butterworth filter.

New questions

Question 1 (easy) [4 marks] A digital lowpass filter must satisfy $0.85 \leq |H(\omega)| \leq 1$ for $0 \leq \omega \leq 0.2\pi$ and $|H(\omega)| \leq 0.15$ for $0.5\pi \leq \omega \leq \pi$. It is to be designed with the bilinear transform ($T = 1$) from an analogue Butterworth filter.

a. Write down the analogue specifications. [2]
b. Find the minimum order N and the range of Ω_c that meets both specs. [2]

Question 2 (easy–medium) [5 marks] A first-order digital lowpass filter with its 3-dB cutoff at 1 kHz is required. $F_s = 8$ kHz. Use the analogue prototype $H_a(s) = \dfrac{\Omega_c}{s + \Omega_c}$ and the bilinear transform with $T = 1/F_s$.

a. Find the prewarped Ω_c in rad/s. [1]
b. Find $H(z)$ and the difference equation. [3]
c. A classmate skips prewarping and uses $\Omega_c = 2\pi(1000)$ rad/s. At what frequency (Hz) is their digital filter 3 dB down? [1]

Question 3 (medium) [5 marks] An analogue 4th-order Butterworth lowpass filter has its poles on a circle of radius $2\pi(500)$ rad/s.

a. Give the four poles of $H(s)$. [2]
b. Write $H(s)$ as a product of two real quadratics. [1]
c. Find the attenuation at 1.5 kHz. [1]
d. What minimum order would give at least 60 dB attenuation at 1.5 kHz with the same pole radius? [1]

Question 4 (medium–hard) [8 marks] — full design A, stopband met exactly A digital lowpass filter is required with passband ripple ≤ 1 dB up to 1 kHz, stopband attenuation ≥ 20 dB from 3 kHz, and $F_s = 8$ kHz. Design it by applying the bilinear transform ($T = 1$) to an analogue Butterworth filter, choosing Ω_c to meet the **stopband** exactly (as in T9 Q5).

a. Find the prewarped edges and the order N. [2]
b. Find Ω_c and $H(s)$. [2]
c. Find $H(z)$ and the difference equation. [3]
d. Check the design against the specs. [1]

Question 5 (hard, multi-part) [10 marks] — full design B, passband met exactly A digital lowpass filter is required with passband ripple ≤ 1 dB up to 2 kHz, stopband attenuation ≥ 20 dB from 4 kHz, and $F_s = 12$ kHz. Use the bilinear transform on an analogue Butterworth filter, with Ω_c chosen to meet the **passband** exactly (L10 slide 17).

a. Find ω_p, ω_s, the prewarped edges ($T = 1$) and N. [2]
b. Find Ω_c, the LHP poles and $H(s)$. [2]
c. Find $H(z)$ and the difference equation. [3]
d. Verify the specs and the stability. [2]
e. Show that using $T = 1/F_s$ instead of $T = 1$ gives the same $H(z)$. At what frequency in Hz is the digital filter 3 dB down? [1]

Question 6 (hard, trap-finder) [4 marks] For the specification in Question 5, a student skips prewarping. They use $\Omega_p = \omega_p = \pi/3$ and $\Omega_s = \omega_s = 2\pi/3$ directly ($T = 1$), then apply the bilinear transform.

a. What order does their calculation give? [1]
b. Suppose they keep $N = 3$ and set Ω_c to meet their (unwarped) passband exactly. Find the actual digital gain at ω_p and ω_s. Does the design meet the specs? [3]

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6. Quick recall check

1. Under the bilinear transform, where do $s = 0$ and $s = \infty$ go in the z-plane?
2. Write the prewarping formula, and say which spec quantities it changes.
3. A passband ripple of 1 dB means $\delta_p = ?$ In the order formula, what is $(1 - \delta_p^2)/\delta_p^2$?
4. The Butterworth poles of $H(s)$ lie on what circle, and at what angles for $N = 3$?
5. The order formula gives $N = 3.02$. What order do you use, and why?